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SERIES OF TECHNICAL NOTES ON THE IMPACT OF NATURAL GAS DEFICIT AND THE INCREASE IN PRICES FOR END USERS: GENERAL PRESENTATION

Note 2. Fiscal cost of increased subsidies due to rising natural gas prices

Authors:

Sergio Cabrales

Juan Benavides

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FEDESARROLLO
Centro de Investigación Económica y Social

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Technical notes series on the impact of the natural gas shortage and price increases for end users: general overview

Due to a combination of past and present public policies, which have introduced transaction costs to the exploration and exploitation of conventional and unconventional domestic natural gas, indecisions on the use, size, location and remuneration of new import facilities, inflexibilities in contracting and charging by distance of the transport system, Colombia faces a deficit and possible shortage of natural gas in the short term.

Fedesarrollo presents four technical notes on the impacts of the natural gas deficit due to the reduction in domestic supply, the difficulties in expanding import capacities in the short term, and the increase in prices for end users due to import needs.
These notes aim to provide elements for decision-making within the State and to inform the public with public data and arguments guided by the general interest.

The titles of the technical notes are as follows:

Note 1. Increase in the price of residential natural gas in scenarios of greater imports.

Note 2. Fiscal cost of increased subsidies due to rising natural gas prices.

Note 3. Macroeconomic costs of reducing the supply of natural gas.

Note 4. Short and medium term public policy and regulation.

Note 5. Energy poverty due to reduced supply and increased price of natural gas.

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Note 2. Fiscal cost of increased subsidies due to rising natural gas prices

Sergio Cabrales and Juan Benavides (Fedesarrollo, Colombia)1 .

Objective and main message

Over the past twelve years, Colombia's proven natural gas reserves have decreased by more than sixty percent, falling from 5,727 gigacubic feet (GCF) in 2012 to 2,064 GCF in 2024. Projections of the supply and demand balance, based on official information, indicate that by the end of 2026, the supply of domestic and imported natural gas will not be sufficient to meet the demand projected by the UPME (Mining and Energy Planning Unit). Imported gas, in addition to being insufficient to address the local deficit due to the size of the regasification plant in Cartagena, has prices that can be up to three times higher than those of domestic gas.

The number of regulated natural gas users in Colombia grew by 218.7% between 2005 and 2024, increasing from 3.6 million to 11.5 million, with an average of 420,000 new users annually, primarily residential. According to projections by DANE (National Administrative Department of Statistics) for 2025, 78% of households in municipal capitals (approximately 11.5 million) are subscribed to natural gas service, used mainly for cooking, water heating, and, to a lesser extent, space heating. The distribution by socioeconomic stratum shows that 59.8% of users belong to strata 1 and 2, who receive subsidies of 60% and 50%, respectively, while strata 5 and 6 (5.3% of the total) contribute 20% to finance these subsidies.

The average monthly consumption of natural gas in Colombian households is 12.8 m³, representing 17.9% of national demand. Bogotá leads with 15 m³/month, followed by Bucaramanga (13.5 m³/month), Barranquilla (13.4 m³/month), Medellín (11.9 m³/month), and Cali (9.5 m³/month), where the lower consumption is explained by higher gas prices. By socioeconomic stratum, a U-shaped pattern is observed: strata 1 and 2 consume more due to subsidies, strata 3 and 4 consume less, and strata 5 and 6 show a significant increase (up to 22.6 m³/month in stratum 6) due to their greater purchasing power and additional uses of gas.

This Note 2 in the series estimates the fiscal cost of higher natural gas prices under conditions of domestic supply shortages and increasing import needs. The rise in imported gas, which is significantly more expensive, poses challenges such as increased tariffs for regulated users (up to 91.5% in some scenarios).

It is emphasized that the starting point is an unsustainable subsidy scheme inconsistent with poverty reduction in Colombia: between 2005 and 2024, subsidies for strata 1 and 2 grew by 642%, while contributions from strata 5 and 6 only increased by 157%, generating a fiscal deficit that went from COP $0.17 trillion to COP $1.25 trillion.

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[1] Disclaimer: The opinions expressed in this publication are those of the authors and should not be attributed to anyone. Fedesarrollo.

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The Colombian Natural Gas Association, NATURGAS, estimates that 1.3 trillion pesos from the General Budget of the Nation (PGN) are required for 2025 and 1.4 trillion for 2026, in order to cover the deficit generated by the subsidies granted to households in strata 1 and 2.

If half of the supply came from imported gas, the tax burden would worsen and could increase by COP $0.36 trillion annually. In a scenario where the supply basket consists exclusively of imported gas, the increase in this tax burden could reach COP $0.72 trillion annually, underscoring the need to address the subsidy scheme and the decline in domestic supply.

Technical notes 3 and 4 in this series will show the macroeconomic impact of a reduced supply, and on energy poverty.

Context of natural gas use in Colombian homes

Towards the end of the 20th and beginning of the 21st centuries, piped natural gas and LPG achieved widespread penetration in urban areas and, to a lesser extent, in rural areas. This progress resulted from energy substitution policies focused on replacing more polluting and dangerous fuels. A crucial aspect was the gradual eradication of highly hazardous fuels such as kerosene (industrial alcohols) and their replacement with gas, especially in the most vulnerable socioeconomic strata. Likewise, the substitution of liquid fuels and coal with gas was promoted in the industrial sector. Gas has been recognized as an energy source with a progressive social impact and a factor supporting productivity in Colombia.

The two main challenges for gas users are security of supply —due to the decline in domestic supply and the lack of import infrastructure— and the increase in tariffs, resulting from the higher cost of gas molecules when imported and the possible increase in transportation costs.

In Colombia, the decline in domestic natural gas supply is due to a combination of structural, fiscal, operational, and social factors. These include high exploration risk with a low success rate (23%), the increased *government take* following the 2022 tax reform, which raised the tax burden from 63% to 80% in 2022, and a drastic reduction in exploration activity. Added to this are regulatory obstacles related to obtaining environmental and social licenses, as well as growing insecurity that has led to attacks and blockades of oil infrastructure.

Regulated gas users

Regulated users are those whose natural gas supply contracts are subject to the uniform conditions established by the public utility companies and regulated by the CREG (Energy and Gas Regulatory Commission). They generally have consumption levels of less than 100,000 cubic feet per day (CFPD), such as residential, small commercial, or low-demand industrial users. In Colombia, natural gas is used in homes for cooking, water heating, and, to a lesser extent, space heating.

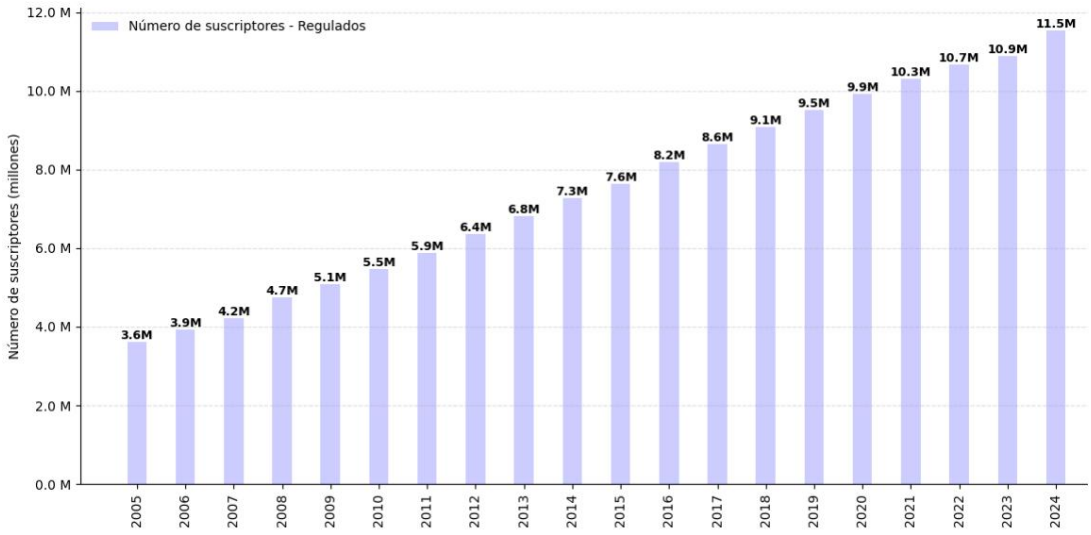
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Law 142 of 1994, developed through multiple resolutions issued by the Energy and Gas Regulatory Commission (CREG), determines: (i) the definition of subsistence consumption (CREG Resolution 057 of 1996); (ii) the contributions according to user type (CREG Resolutions 124 and 015 of 1996 and 1997); (iii) the criteria for remunerating distribution and marketing activities (CREG Resolutions 202 of 2013 and 102 003 of 2022); and (iv) the formulas for general tariffs for regulated users (CREG Resolution 137 of 2013).

Over the past two decades, the number of regulated users has grown by 218.7%, from 3.6 million in 2005 to more than 11.5 million in 2024, as shown in Figure 1. During this period, an average of 420,000 new regulated users, mostly residential, have been connected annually. Based on the 2018 National Population and Housing Census (CNPV), DANE projects that there will be 14.7 million households in municipal capitals by 2025. Therefore, it is estimated that approximately 78% of these households are subscribed to gas service.
natural.

Only 3.2 million households remain to subscribe to the service, which would limit subscriber growth in the coming years. However, the challenge of retaining the current 11.5 million users in the face of potential price increases also remains.
after the loss of self-sufficiency of essential demand.

Chart 1. Number of subscribers - Regulated



Source: Fedesarrollo's own elaboration based on data from the Single Information System (SUI).

Stratum of households that use gas

Of the more than 11.5 million households in Colombia that use gas, primarily for cooking, 24.2% and 35.6% belong to socioeconomic strata 1 and 2, respectively, meaning that 3 out of every 5 households using gas are in these two strata (Figure 2). These households receive a subsidy of 60% if they belong to stratum 1 and 50% if they belong to stratum 2, applied to their total gas bill, which includes variable costs that depend on gas consumption and the

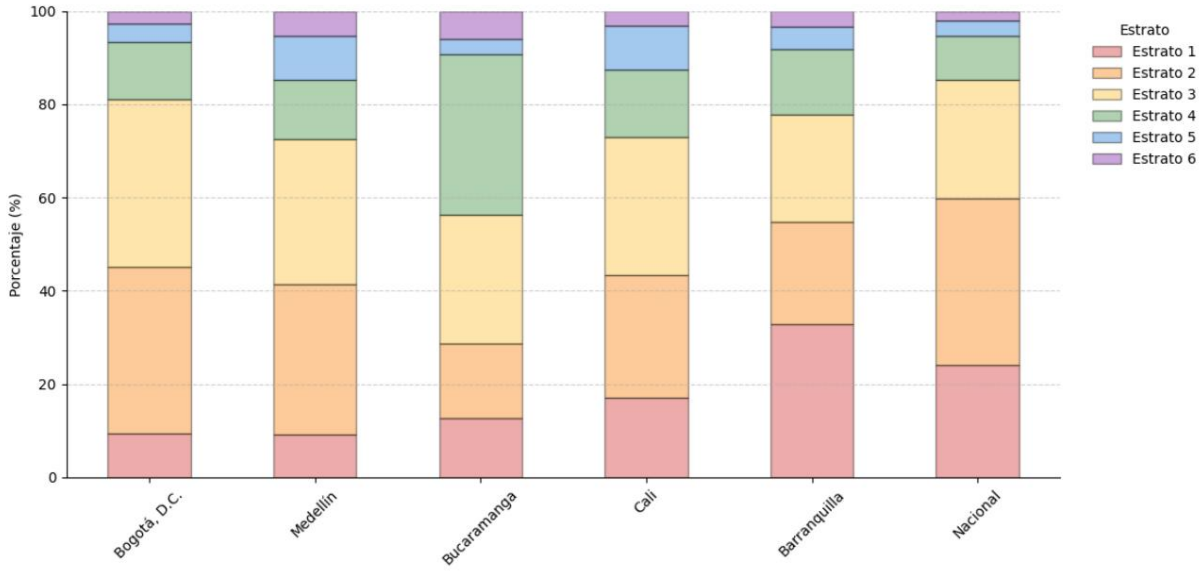
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Billing cost. 2 With the intention of preventing excessive household consumption

For those belonging to strata 1 and 2, subsidies apply to consumption below 20 cubic meters (m3) per month, which is called basic or subsistence consumption.3

25.4% and 9.4% of households using gas belong to socioeconomic strata 3 and 4, respectively, meaning they do not receive a subsidy for gas service. Unlike gas for households in stratum 3, the electricity rate for this stratum does include a 15% subsidy. Finally, 3.4% and 1.9% of households using gas belong to strata 5 and 6, respectively; their rates include a contribution equivalent to 20% of the total billed.

Chart 2. Distribution by subscriber stratum - residential users (%)



Source: Fedesarrollo's own elaboration based on data from the Single Information System (SUI).

Among the five main cities analyzed, Barranquilla has the highest percentage of households in socioeconomic stratum 1 that use gas, at 32.9%, meaning that one in three households belongs to this stratum. In contrast, in cities like Bogotá and Medellín, this percentage is less than 10%, while Bucaramanga and Cali have intermediate values, at 12.6% and 17.1%, respectively. Bogotá and Medellín account for 35.8% and 32.1% of households in socioeconomic stratum 2, respectively, which pay only half of the gas bill. Meanwhile, Bucaramanga has the lowest percentage of households in stratum 2, at 16%, while Cali and Barranquilla register values of 26.3% and 21.9%, respectively.

Regarding contributions, in these five cities between 8% and 15% of households belong to strata 5 and 6, which contribute 20% of the value of their bill to subsidize

2 CREG Resolution 186 of December 30, 2010, modified by CREG Resolution 186 of 2013 and this has been extended by CREG Resolutions 186 of 2014, 241 of 2015, 152 of 2018, 198 of 2019, 003 of 2021, 214 of 2021 and 105 005 of 2022.
3 CREG Resolution 057 of 1996.

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Gas consumption by households in strata 1 and 2. Medellín has the highest percentage of households that contribute (14.7%), while Barranquilla registers the lowest (8.4%).

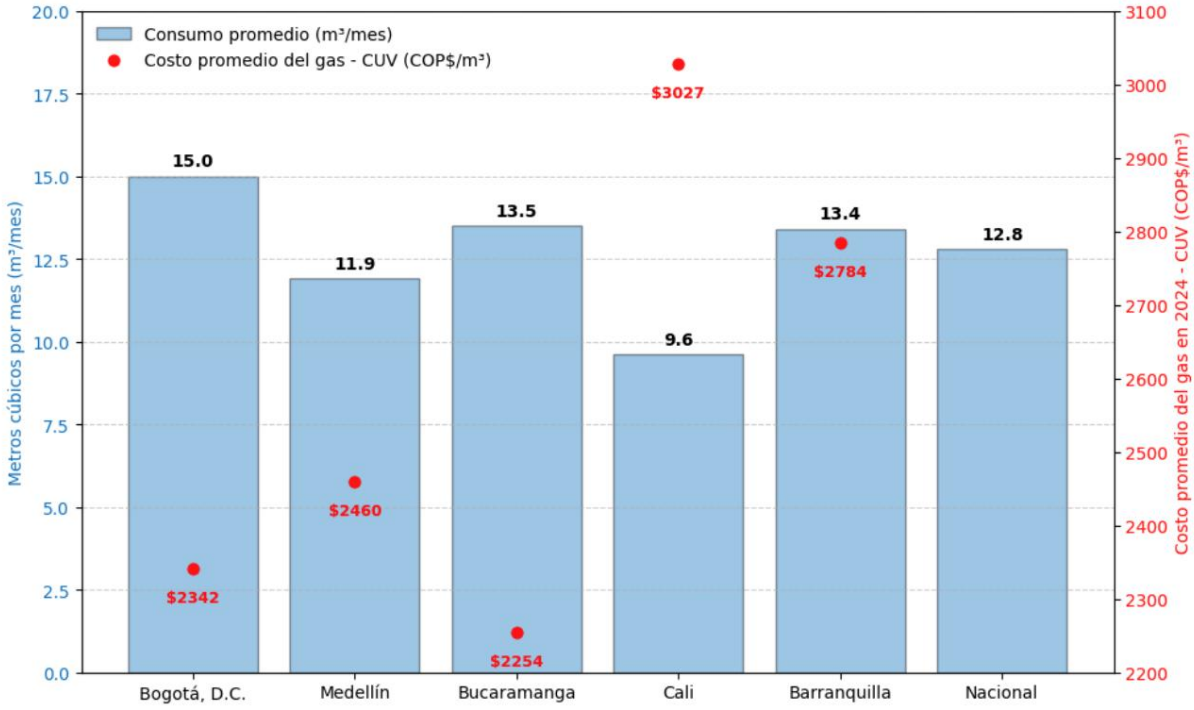
In summary, at the national level:

- Of every five households that use gas, three receive a subsidy of between 50% and 60% of the cost total of your bill.
- Only 5% of households make a contribution equivalent to 20%, which forces the national government to cover the difference between subsidies and contributions, generating a tax burden significant.

Household gas consumption

Nationally, the average monthly consumption of the residential sector is 12.8 cubic meters per household (m3 / month), as shown in Figure 3. This aggregate consumption represents 17.9% of the national total, surpassed only by the demand from thermal power plants (31.4%) and the industrial sector (30.7%). 4

Chart 3. Average monthly consumption of natural gas (m³/month)



Source: Fedesarrollo's own elaboration based on data from the Single Information System (SUI).

Households in Bogotá consume an average of 15 cubic meters per month (m³/month), exceeding the national average (12.8 m³/month) and that of the other four cities analyzed, as shown in

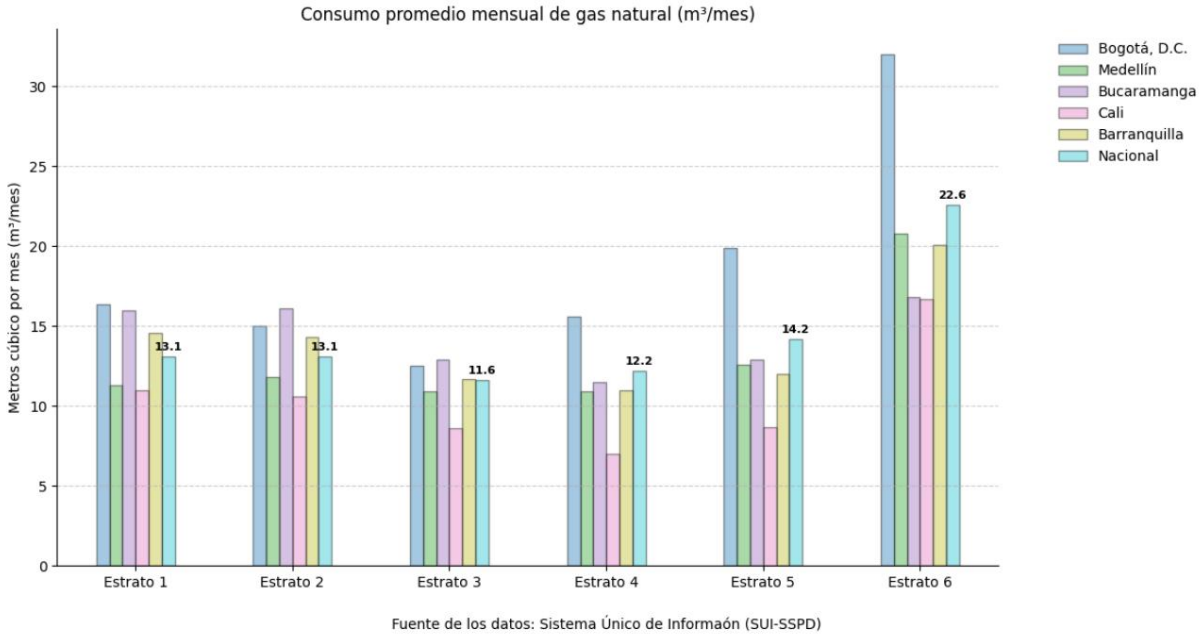
4 Cabrales, S. & Benavides J. (2025). Series of technical notes on the impact of the natural gas deficit and the increase in prices for end users: general presentation. Note 1. Increase in the price of residential natural gas in scenarios of greater imports. Bogotá Fedesarrollo. 14 p.

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Graph 3. This is partly explained by the relatively low price of gas and its additional use, besides cooking food, in activities such as water heating and space heating, due to the lower temperatures.

In Medellín, with 11.9 m³/month; Bucaramanga, with 13.5 m³/month; and Barranquilla, with 13.4 m³/month, households register gas consumption close to the national average. The case of Cali stands out with 9.5 m³/month, where consumption is 25% lower than the national average, mainly due to having the highest gas price among the five cities analyzed.

Chart 4. Average monthly natural gas consumption by socioeconomic stratum (m³/month)



Source: Fedesarrollo's own elaboration based on data from the Single Information System (SUI).

Analyzing gas consumption in these five cities and the national average by socioeconomic stratum reveals a U-shaped pattern (Figure 4). Households in strata 1 and 2 consume more gas on average than those in strata 3 and 4, which is explained by the existence of subsidies of between 50% and 60% of the total bill. On the other hand, households in stratum 5 (14.2 m³/month) and stratum 6 (22.6 m³/month) show a significant increase in consumption compared to the other strata, due to their greater purchasing power and the additional use of gas beyond cooking, as is the case in Bogotá.

Balance between subsidies and contributions for regulated users

The natural gas tariff for regulated users, known as the Unit Cost of Provision of Service (CU), is composed of: (i) the product between consumption in cubic meters (m³/month) and the Variable Unit Cost (VUC) that includes the cost components of gas molecules (G),

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transport (T) and distribution (D); and (ii) the Fixed Unit Cost (FUC), which corresponds to the fixed component charged on each invoice, regardless of the volume consumed.5

The CUV varies depending on the origin of the gas (domestic or imported), the transport from the source of production or import to the distributor *(city gate),* and the distributor, which represents the part of the tariff that depends directly on the user's consumption level.

As mentioned, users in strata 1 and 2 can receive subsidies of 60% and 50%, respectively, while users in strata 5 and 6, as well as some commercial and industrial sectors with consumption below 100,000 cubic feet per day (CFD), contribute 20% to finance these subsidies.

In 2005, subsidies (red bars) for households in strata 1 and 2 represented COP $0.19 trillion, compared to COP $1.41 trillion today, which is equivalent to an increase of 642% in the last 20 years, as shown in Graph 5. This increase is explained by the growth in the number of users, which went from 3.6 million in 2005 to more than 11.5 million in 2024, of which about 60% belong to strata 1 and 2.

Chart 5. Balance between contributions and subsidies among regulated users (2025 pesos)



Source: Fedesarrollo's own elaboration based on data from the Single Information System (SUI).

To compensate for these subsidies, contributions (green bars) from strata 5 and 6, and small merchants and industrialists are used, which correspond to 20% of the total value of the bill. However, these contributions increased from COP $0.07 trillion in 2005 to COP $0.18 trillion in 2024, representing an increase of 157%, much lower than the growth of subsidies.

5 = (+)/(1 ÿ)+ (ÿ calorific.) where are the recognized losses and It is the power factor

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This structural imbalance between subsidies and contributions has led to the fiscal burden for the Government (black spots) going from COP $0.17 trillion in 2005 to around COP $1.25 trillion in recent years, representing an increase of more than one trillion pesos in 20 years.

Unregulated users

Non-regulated users are those who, due to their high gas consumption, can directly negotiate supply terms with gas marketers or producers, without being subject to tariffs regulated by the CREG (Energy and Gas Regulatory Commission). They can enter into bilateral contracts with marketers, producers, or transporters, allowing them to negotiate specific prices and conditions, as established in Article 22 of Law 2128 of 2021. They are not subject to the uniform terms of residential public utility contracts, which gives them greater flexibility in contracting. These are typically large industries that require gas for their industrial processes, such as refineries, cement plants, steel mills, petrochemical plants, power generation plants, or companies with self-generation of electricity that require large volumes of gas.

The number of unregulated subscribers increased from 502 in 2005 to 1,162 in 2024, a rise of only 660 subscribers over a 20-year period (Figure 6). The highest number of unregulated subscribers was reached in 2022, with a total of 1,847, but this figure decreased to 1,134 in 2023.

Chart 6. Number of subscribers - Unregulated

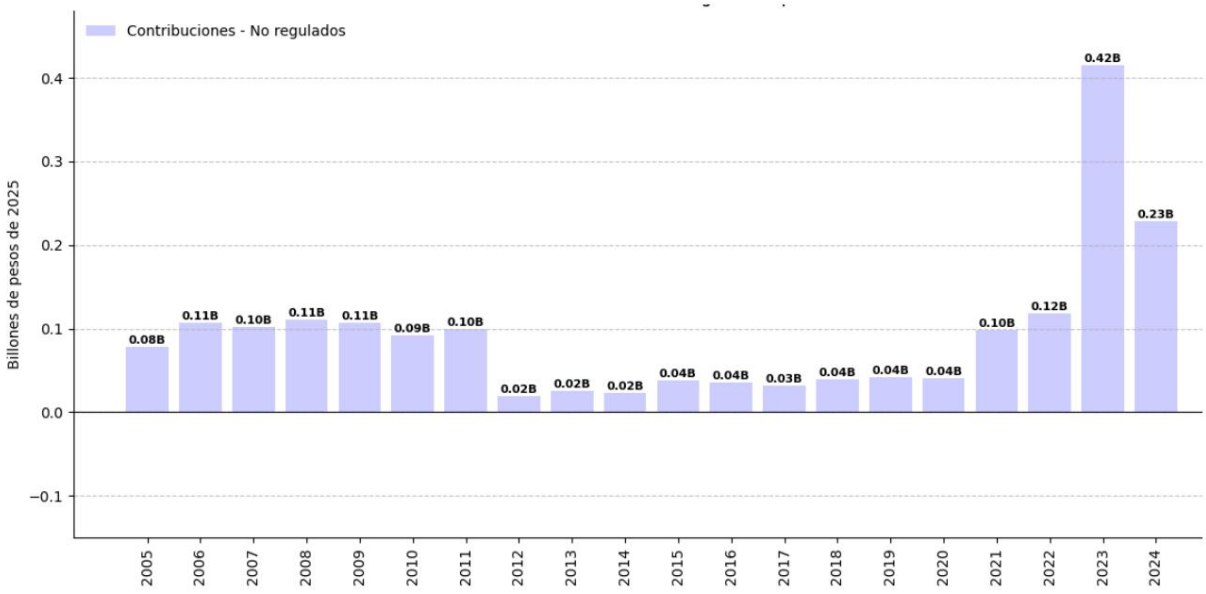


Source: Fedesarrollo's own elaboration based on data from the Single Information System (SUI).

Some commercial and industrial users, such as thermal power plants, refineries, the health sector, and the education and social assistance sectors, are exempt from the 20% contribution on total billing. Figure 7 shows the contributions from unregulated users who are not exempt, which have ranged between COP $0.02 trillion and COP $0.42 trillion over the last 20 years. This also contributes, to some extent, to reducing the tax burden associated with subsidies for socioeconomic levels 1 and 2.

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Chart 7. Contributions from unregulated users (2025 pesos)



Source: Fedesarrollo's own elaboration based on data from the Single Information System (SUI).

Overall balance between subsidies and contributions

After adding the contributions of regulated and unregulated users, the overall balance between subsidies and contributions for the entire system is presented in Chart 8.

Chart 8. Overall balance between contributions and subsidies (2025 pesos)



Source: Fedesarrollo's own elaboration based on data from the Single Information System (SUI).

The annual difference between subsidies and contributions has increased from COP $0.05 trillion in 2005 to around COP $1 trillion today, representing a drastic increase in the fiscal cost for the

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nation. This increase, which is not fiscally sustainable, is mainly explained by the growth in the number of households in strata 1 and 2 that use gas in the last two decades.

Over the past five years, the net annual deficit between subsidies and contributions has fluctuated between COP $0.8 trillion and COP $1.1 trillion. This difference must be covered by the national government to maintain the subsidy structure for socioeconomic levels 1 and 2.

Increased gap between subsidies and contributions

The price of imported gas—in the form of liquefied natural gas (LNG)—is usually indexed to international benchmarks such as the *Henry Hub* (United States). Added to this price are the costs of liquefaction, maritime transport, regasification at import terminals, and port charges. As a result, the price of imported gas is higher than that of domestically produced gas.

The price of natural gas in Colombia depends on the origin of the supply (whether domestic production or imports), contractual conditions, and market dynamics.
national and international. Before 2023, the average prices of domestic gas contracts ranged between USD $2.80 and USD $5.40 per million BTU (MBTU).

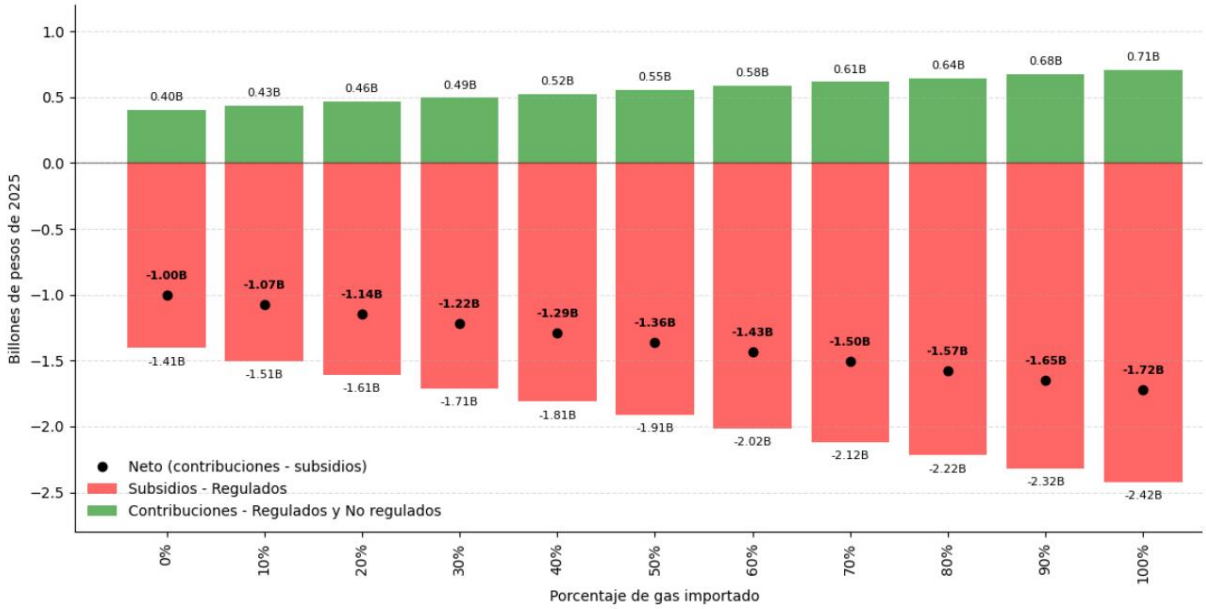
In 2023, these prices rose to an average of USD $7.50 per MBTU, and have recently exceeded USD $12.00 per MBTU. Meanwhile, the average price of imported gas has ranged between USD $14.96 and USD $18.39 per MBTU, significantly higher than the price of domestic gas. This difference is mainly due to the additional costs of liquefaction, shipping, and regasification, as well as the purchase of gas on the secondary market, given the lack of long-term contracts in the international primary market.

Note 1 estimated that if half of the supply came from imported gas, the tariff increase for regulated users could reach 44.6% in Bogotá, 44.9% in Medellín, 45.8% in Bucaramanga, 30.3% in Cali, and 11.7% in Barranquilla, respectively. In a scenario where the supply consists exclusively of imported gas, the increases could be 89.3% in Bogotá, 89.7% in Medellín, 91.5% in Bucaramanga, 60.7% in Cali, and 23.4% in Barranquilla.

Based on the previous results, it was estimated that if half of the supply came from imported gas, the annual tax burden would increase by COP $0.36 trillion, rising from COP $1 trillion to COP $1.36 trillion (Figure 9). However, if 100% of the gas supply were imported, the increase in the tax burden would be approximately COP $0.72 trillion per year. These estimates do not consider a possible increase in the number of households in socioeconomic strata 1 and 2, nor variations in consumption resulting from increases in gas prices.

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Chart 9. Variation of annual subsidies and contributions according to the percentage of imports (trillions of pesos 2025)



Source: Prepared by Fedesarrollo with data from the Single Information System (SUI), Superintendence of Public Utilities, Electronic Gas System (SEGAS) and Energy and Gas Regulation Commission (CREG).

Final considerations

The Colombian Natural Gas Association, NATURGAS, estimates that 1.3 trillion pesos from the General Budget of the Nation (PGN) are required for 2025 and 1.4 trillion for 2026, in order to cover the deficit generated by the subsidies granted to households in strata 1 and 2.

The increase in the cost of subsidies due to higher costs from the increase in imported gas, which could reach COP$0.36 trillion with 50% imports, comes in a context of total tax revenue falling 4.36% between 2023 and 2024, in contrast to a 6.14% increase in public spending in the same period, and government delays in paying subsidies to companies in the sector, despite the progress made in May 2025.

The government must not only pay the outstanding balance for subsidies to gas companies, but also move towards a final solution to avoid perpetuating the unsustainable deficit between contributions and subsidies.